

An Overview and Deep Investigation on Sampled-Data-Based Event-Triggered Control and Filtering for Networked Systems

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Abstract—This paper provides an overview and makes a deep investigation on sampled-data-based event-triggered control and filtering for networked systems. Compared with some existing event-triggered and self-triggered schemes, a sampled-data-based event-triggered scheme can ensure a positive minimum inter-event time and make it possible to jointly design suitable feedback controllers and event-triggered threshold parameters. Thus, more attention has been paid to the sampled-data-based event-triggered scheme. A deep investigation is first made on the sampled-data-based event-triggered scheme. Then, recent results on sampled-data-based event-triggered state feedback control, dynamic output feedback control, H_∞ filtering for networked systems are surveyed and analyzed. An overview on sampled-data-based event-triggered consensus for distributed multiagent systems is given. Finally, some challenging issues are addressed to direct the future research.

Index Terms—Event-triggered control (ETC), event-triggered filtering, networked systems, sampled-data-based event-triggered scheme (ETS).

I. INTRODUCTION

IN ADVANCED control applications, control implementation is largely based on a digital platform, where control components (physical plant, sampler, controller, and actuator) are connected through a communication network. In such implementation, a control task consists of sampling physical signals, computing control signals, and implementing actuator signals. Traditionally, control tasks are executed in a periodic way, which leads to so-called time-triggered control. Time-triggered control paves an easy way to design suitable controllers using a fruitful sampled-data system theory [1], [2]. However, in some

cases, where the system reaches its steady state and no disturbance is imposed on the system, executing those control tasks periodically wastes both computation and energy resources of battery-based devices. Moreover, if a sampling period is approaching zero, a large number of redundant sampled signals will be released to a shared communication network with limited bandwidth, which inevitably causes a congested network traffic. Thus, in these cases, time-triggered control should give way to event-triggered control (ETC) [3], [4].

The ETC provides a natural way to execute control tasks, under which, whether or not a control task is executed depends closely on a predefined event-triggered condition, rather than on the lapse of some time period [3], [5]. If an event-triggered condition is violated at some time instant, which means that an event is triggered, current control task should be executed immediately. Compared with a time-triggered control scheme, an ETC scheme can save computation resources, energy resources of battery-based devices, and limited network resources as well. In fact, it has been shown through experimental results [6], [7] that ETC schemes can efficiently reduce the number of control task execution so that communication resources can be saved significantly while retaining satisfactory performance of the resultant closed-loop system. Because of the aforementioned advantages, an idea of ETC has been widely applied to deal with several control and filtering issues, such as state feedback control [8]–[11], output feedback control [5], [12]–[15], passive control [16], tracking control [17], model predictive control [18], and optimal state estimation and H_∞ filtering [19], [20]. Hence, in the last decade, the ETC has become a hot research topic in the field of control and signal processing [21]–[31].

An ETC system consists of two key elements: a feedback controller that computes the control input, and an event-triggered mechanism that determines when control input should be updated [32]. The feedback controller is commonly given *a priori*. The event-triggered mechanism is usually predefined as some condition related closely to *instantaneous* system states or system measurements [3], [8]. Thus, extra hardware is needed to monitor the system states/measurements instantaneously. In order to avoid extra hardware, self-triggered control (STC) is proposed [10]. In STC, the next event time instant is predicted based on previously received data and knowledge on the plant dynamics [33]–[36]. As for more difference between ETC and STC, one can refer to [37] and [38].

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In either ETC or STC, the first important issue is that the minimal time distance between any two consecutive event time instants, which is called the minimum inter-event time, must be strictly greater than zero. If the minimum inter-event time is zero, infinite events occur within a finite time interval, which leads to the so-called Zeno behavior [39]. It is proven [3] that the positive minimum inter-event time is guaranteed to exist for state feedback control of linear physical plants without external disturbance, while it is shown [5] that the positive minimum inter-event time may not exist in the case that an output feedback controller is used in a similar setup. In 2014, properties of inter-event times for ETC systems are investigated in detail [40], and it is proven theoretically that for some classes of systems, even though a positive minimum inter-event time can be ensured in the absence of external disturbance, it possibly becomes zero for arbitrary small external disturbance. Thus, it is not an easy task to ensure that an ETC system has a positive minimum inter-event time especially in the presence of external disturbance, which limits the applications of ETC and STC. Another important issue is the joint design of both the desired feedback controller and the threshold parameter of the event-triggered scheme (ETS). In either ETC or STC, some existing approaches are mostly emulation based [5]. That is, a feedback controller is designed without considering the event-triggered nature of the control system; and the ETS is only designed after the controller is given *a priori*. It is clear to see that, with such an emulation-based approach, it is impossible to obtain an optimal design of the combined feedback controller and the threshold parameter of the event-triggered scheme such that the number of control executions can be minimized while suitable performance for the resultant closed-loop system can be ensured [33].

Bearing the aforementioned two issues in mind, a sampled-data-based ETS is proposed in [41] and further developed in [42]. The main idea of the sampled-data-based ETS lies in three aspects: first, the system measurements are sampled periodically; second, whether or not the sampled data packet should be transmitted is determined by a predefined event-triggered condition, which is related to information on signals only at sampling instants; and third, the resultant closed-loop system is modeled as a time-delay system. It is clear to see that, under the sampled-data-based ETS, extra hardware is not required to monitor the instantaneous system states or measurements; on the other hand, a minimum inter-event time is ensured to be equal to or greater than a sampling period. Another significant feature of the sampled-data-based ETS is that suitable feedback controllers and event-triggered threshold parameters can be jointly designed in terms of a set of linear matrix inequalities. Thus, the proposed sampled-data-based ETS paves an effective way to address the two important issues mentioned previously. Therefore, the sampled-data-based ETS comes to the fore in the field of ETC, and a number of results have been reported in the literature, see, e.g., [43]–[55].

It should be mentioned that, a periodic ETS is proposed in [56] and [57]. Under the periodic ETS, “at each sampling time, state measurements are transmitted over a communication network and the control values are updated only when certain event-triggering conditions are satisfied” [57], but it is assumed that no

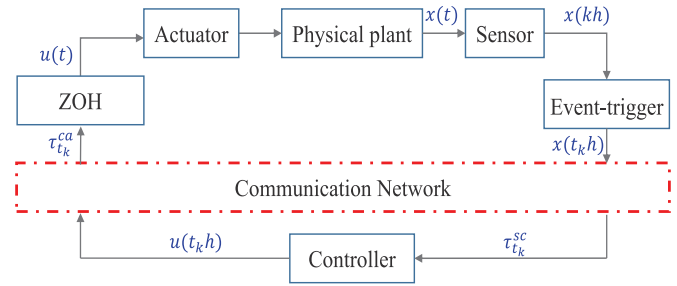


Fig. 1. Simple diagram for an ETC system.

transmission delays, no network-induced delays, and no packet dropouts are encountered when the sampled data packets are transmitted from the physical plant to the controller. Clearly, this assumption is *not* true if the physical plant and the controller are located remotely. Nevertheless, such an assumption does not need in the sampled-data-based ETS. Moreover, although the periodic ETS can ensure a positive minimum inter-event time, “the joint design of the controller and the event-triggered condition is a hard problem” [57]. In fact, under the periodic ETS, the closed-loop system is modeled as an impulsive system [57]. Even though some sufficient conditions on stability and L_2 gain properties are derived using an impulsive system approach, these conditions are dependent closely on solutions to a Riccati differential equation, which makes the controller design more complicated. Furthermore, if both transmission delays and network-induced delays are taken into account, these conditions fail to draw any conclusion on the stability of the closed-loop system, which limits the application scope of the proposed method [57].

This paper provides a survey on recent results of ETC and filtering for networked systems. The emphasis is placed on the sampled-data-based ETS. Section II makes a deep investigation on the sampled-data-based ETS; sampled-data-based ETC is presented in Section III; event-based filtering for networked systems is reviewed in Section IV; a brief overview of event-triggered consensus for distributed multiagent systems (MASs) is presented in Section V; applications of ETC using the sampled-data-based ETS is analyzed in Section VI. Section VII draws the conclusion and poses some challenging problems to direct the future research.

II. DEEP INVESTIGATION OF THE SAMPLED-DATA-BASED ETS

In this section, a deep investigation is made on the sampled-data-based ETS proposed in [41] and [42]. Consider the ETC system shown in Fig. 1, where the physical plant is given by

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(t_0) = \phi_0. \quad (1)$$

For simplicity of presentation, suppose that the state $x(t)$ of the physical plant is available, and is sampled with a period $h > 0$, where h is a constant; and also suppose that the sensor is time driven, while the event trigger, the controller, the zero-order-hold (ZOH), and the actuator are event driven. To proceed with, we start from an event trigger.

A. Event Trigger

The event trigger is used to trigger a new event to occur if a predefined event-triggered condition is violated. The event-triggered condition is defined as

$$e^T(t_k, j)\Omega e(t_k, j) \leq \sigma x^T((t_k + j)h)\Omega x((t_k + j)h) \quad (2)$$

where $e(t_k, j) = x((t_k + j)h) - x(t_k h)$ ($j = 1, 2, \dots$); $t_k h$ stands for the last event time; $\Omega > 0$ is a weighting matrix, and $\sigma > 0$ a threshold parameter. From (2), one can see clearly that each event is triggered at its sampling instant, which means that the minimum inter-event time can be ensured to be equal to or greater than h even though external disturbance is imposed on the physical plant. According to the mechanism of the event trigger, the next event-time instant $t_{k+1}h$ can be determined by

$$t_{k+1}h = t_k h + \min_{j \geq 1} \{jh | e^T(t_k, j)\Omega e(t_k, j) > \sigma x^T((t_k + j)h)\Omega x((t_k + j)h)\}. \quad (3)$$

Thus, we have a time sequence $\{t_1, t_2, \dots, t_k, t_{k+1}, \dots\}$ indicating when events are triggered.

The event-triggered condition (2) is given in the form of the relative error (in the sense of a matrix norm) between $x(t_k h)$ and $x(t_k h + jh)$. Thus, it is reasonable to set the threshold parameter σ to be less than one. The smaller the threshold parameter σ is chosen, the more the events will be triggered. When the threshold parameter σ is sufficiently approaching zero, the event-triggered mechanism reduces to the time-triggered one. As for the weighting matrix Ω , which is inspired by an S -procedure, it aims just to “enhance” the feasibility of the obtained stability criteria [20]. If one modifies the condition (2) as

$$e^T(t_k, j)\Omega_1 e(t_k, j) \leq \sigma x^T((t_k + j)h)\Omega_2 x((t_k + j)h) \quad (4)$$

where $\Omega_1 > 0$ and $\Omega_2 > 0$, the feasibility of the derived stability criteria will be further “enhanced,” but it may be potential that less network resources may be saved due to that one may find two matrices Ω_1 and Ω_2 such that the condition (4) is violated. Moreover, the threshold parameter σ can be absorbed by Ω_2 . In this situation, how Ω_1 and Ω_2 affect both the performance of an event-triggered system and the network resource saving needs to be further investigated.

B. Expression of Control Input $u(t)$

Suppose that the triggered data packets $(t_k, x(t_k h))$ ($k = 1, 2, \dots$) can be successfully transmitted to the ZOH through the communication network subject to network-induced delays (see Fig. 1) $\tau_{t_k} = \tau_{t_k}^{sc} + \tau_{t_k}^{ca} \leq \bar{\tau} < \infty$, where $\bar{\tau}$ is a known constant. Then, under a state feedback controller, the control input signal $u(t)$ can be expressed as

$$u(t) = u(t_k h) = Kx(t_k h), \quad t \in [t_k h + \tau_{t_k}, t_{k+1}h + \tau_{t_{k+1}}). \quad (5)$$

This expression is quite similar to the one in [58], where no sampled-data-based ETS is used. If one sets $d(t) = t - t_k h$, one has that $u(t) = Kx(t - d(t))$. Thus, a delay system model

can be obtained for the ETC system. However, this model is not beneficial to stability analysis and control synthesis of ETC systems since the time delay $d(t)$ may be very large or even infinity, especially in the case where the event-triggered condition (2) is always satisfied after the time instant $t_k h$.

One key contribution of Yue *et al.* (2013) [42] is an introduction of a piecewise function $d(t)$ defined on $\mathcal{I}_{t_k} := [t_k h + \tau_{t_k}, t_{k+1}h + \tau_{t_{k+1}})$. The main idea is to decompose the time interval $\mathcal{I}_{t_k}$ into some subintervals such that $\mathcal{I}_{t_k} = \bigcup_{i=0}^{\rho_{t_k}} \mathcal{I}_i$, where

$$\mathcal{I}_i = \begin{cases} [t_k h + \tau_{t_k}, t_k h + h + \bar{\tau}), & i = 0 \\ [t_k h + ih + \bar{\tau}, t_k h + (i+1)h + \bar{\tau}), & i = 1, \dots, \rho_{t_k} - 1 \\ [t_k h + \rho_{t_k} h + \bar{\tau}, t_{k+1}h + \tau_{t_{k+1}}), & i = \rho_{t_k} \end{cases} \quad (6)$$

and $\rho_{t_k} \geq 0$ is an integer satisfying

$$t_k h + \rho_{t_k} h + \bar{\tau} \leq t_{k+1}h + \tau_{t_{k+1}} < t_k h + (\rho_{t_k} + 1)h + \bar{\tau}.$$

In this situation, a piecewise function $\tilde{d}(t)$ is defined as

$$\tilde{d}(t) := \begin{cases} t - t_k h, & t \in \mathcal{I}_0 \\ t - t_k h - jh, & t \in \mathcal{I}_j, j = 1, \dots, \rho_{t_k} - 1 \\ t - t_k h - \rho_{t_k} h, & t \in \mathcal{I}_{\rho_{t_k}} \end{cases} \quad (7)$$

which satisfies

$$\underline{\tau} := \min_{k \geq 1} \{\tau_{t_k}\} \leq \tilde{d}(t) < h + \bar{\tau}. \quad (8)$$

Moreover, introduce another piecewise function $\tilde{\eta}(t)$ as

$$\tilde{\eta}(t) := \begin{cases} x(t_k h) - x(t_k h), & t \in \mathcal{I}_0 \\ x(t_k h) - x(t_k h + jh), & t \in \mathcal{I}_j, j = 1, \dots, \rho_{t_k} - 1 \\ x(t_k h) - x(t_k h + \rho_{t_k} h), & t \in \mathcal{I}_{\rho_{t_k}}. \end{cases}$$

Then, $\tilde{\eta}(t) = x(t_k h) - x(t - \tilde{d}(t))$ ($t \in \mathcal{I}_{t_k}$), which leads to

$$u(t) = Kx(t_k h) = K[\tilde{\eta}(t) + x(t - \tilde{d}(t))], \quad t \in \mathcal{I}_{t_k}. \quad (9)$$

By the definition of $\tilde{\eta}(t)$, the event-triggered condition (2) can be rewritten as, for $t \in \mathcal{I}_{t_k}$

$$\tilde{\eta}^T(t)\Omega\tilde{\eta}(t) \leq \sigma[\tilde{\eta}(t) + x(t - \tilde{d}(t))]^T\Omega[\tilde{\eta}(t) + x(t - \tilde{d}(t))]. \quad (10)$$

The expression (9) of $u(t)$ plays a key role in the sampled-data-based ETS [42], which allows us to establish a useful time-delay model for an ETC system. It is worth pointing out that there is a simpler way to derive the aforementioned expression of $u(t)$. In fact, denote $d_k := t_{k+1} - t_k$. Then, the interval $[t_k h, t_{k+1}h)$ can be decomposed into the union of d_k sampling intervals $[t_k h, t_{k+1}h) = \bigcup_{j=1}^{d_k} \mathcal{J}_j$, where $\mathcal{J}_j = [t_k h + (j-1)h, t_k h + jh)$. If we “artificially” introduce some scalars $\tau_{t_k j}$ such that

$$t_k h + (j-1)h + \tau_{t_k(j-1)} \leq t_k h + jh + \tau_{t_k j}$$

$$\underline{\tau} \leq \tau_{t_k j} \leq \bar{\tau}, \quad \tau_{t_k 0} = \tau_{t_k}, \quad \tau_{t_k d_k} = \tau_{t_{k+1}}$$

the time interval $\mathcal{I}_{t_k}$ can be decomposed as $\mathcal{I}_{t_k} = \cup_{j=1}^{d_k} \Pi_j$ where $\Pi_j := [t_k h + (j-1)h + \tau_{t_k(j-1)}, t_k h + jh + \tau_{t_k j})$. Thus, one can define two piecewise functions $d(t)$ and $\eta(t)$ as

$$\begin{cases} d(t) = t - t_k h - (j-1)h \\ \eta(t) = x(t_k h) - x(t_k h + (j-1)h), \quad t \in \Pi_j. \end{cases} \quad (11)$$

From the aforementioned analysis, the simple decomposition of the interval $\mathcal{I}_{t_k}$ leads to simple forms of $d(t)$ and $\eta(t)$. As a result, the control input signal $u(t)$ can be expressed as

$$u(t) = K[\eta(t) + x(t - d(t))], \quad t \in \Pi_j \quad (12)$$

which is similar to (9). Moreover, by the definition of $\eta(t)$, the event-triggered condition (2) can be rewritten as, for $t \in \Pi_j$

$$\eta^T(t) \Omega \eta(t) \leq \sigma[\eta(t) + x(t - d(t))]^T \Omega [\eta(t) + x(t - d(t))]$$

which is similar to (10).

C. Time-Delay System Model

Substituting (12) into (1), we have the closed-loop system

$$\begin{cases} \dot{x}(t) = Ax(t) + BKx(t - d(t)) + BK\eta(t) \\ x(\theta) = \phi(\theta), \quad \theta \in [-h - \bar{\tau}, 0], \quad t \in \Pi_j \end{cases} \quad (13)$$

where $\phi(\theta)$ is the supplemented initial condition. The system (14) presents a novel time-delay model for the ETC system under the event-triggered condition (2). Notice that the time-delay $d(t)$ belongs to the interval $[\underline{\tau}, h + \bar{\tau})$, which depends only on the sampling period h and the lower and upper bounds of network-induced delays. With this model, it is easy to investigate the effects of network-induced delays and the sampling period on the ETC. Furthermore, this model can be regarded as a time-delay system with an external disturbance $\eta(t)$. Thus, the fruitful theory on both time-delay systems and perturbed differential equations with retarded arguments can be applied to deal with the problem of ETC for networked systems.

D. Some Comments on the Sampled-Data-Based ETS

The sampled-data-based ETS gives a novel way to model an ETC system as a time-delay system. More significantly, this model makes it possible to jointly design suitable event-triggered controllers and event-triggered parameters in terms of linear matrix inequalities rather than using an emulation-based approach [3], [5], [17]. However, the sampled-data-based ETS has some limitations as follows.

- 1) Since there is a need to check the event-triggered condition (2) at every sampling instant, the computation burden is increased largely, especially for a small sampling period $h > 0$.
- 2) If the sensor in Fig. 1 is a battery-based device, the sampled-data-based ETS cannot save any energy resources for such device. On the contrary, more energy resources will be consumed if the sampling period h is very small.
- 3) The drawback of the periodic sampling is completely inherited. For example, it is possible that some necessary signals cannot be sampled, especially when the signals are fluctuating between a large range.

- 4) With the event-triggered condition (2), there may exist a time instant, say $t_f < \infty$, such that no any events will be triggered for $t > t_f$ if the system reaches its steady state and no disturbance is imposed on the system.
- 5) The time-delay system model (14) does not take data packet dropouts into account, which means that good quality of service should be ensured for the communication network. If the data packet triggered from the event trigger is lost during the transmission through the communication network, the aforementioned time-delay system model (14) subject to (10) will be not available since the extra constraint (10) may be breached.

III. SAMPLED-DATA-BASED ETC

In this section, we provide an overview on recent results of ETC using the sampled-data-based ETS, including state feedback control and dynamic output feedback (DOF) control. We analyze these results deeply.

A. Sampled-Data-Based Event-Triggered State Feedback Control

Based on the time-delay system model (14) subject to (10), H_∞ state feedback control is investigated in [42], where the following Lyapunov–Krasovskii functional (LKF) is employed

$$\begin{aligned} V_n(t) = & x^T(t)Px(t) + \int_{t-\tau_M}^t x^T(s)Qx(s)ds \\ & + \int_{-\tau_M}^0 \int_{t+\theta}^t \dot{x}^T(s)R\dot{x}(s)dsd\theta \end{aligned} \quad (14)$$

where $\tau_M = h + \bar{\tau}$. By using the LKF method, some sufficient condition on the bounded realness of the closed-loop system is derived [42], from which

- 1) one can derive the admissible maximum upper bound $\bar{\tau}$ for given values of h, σ , and H_∞ performance $\gamma > 0$, which indicates the maximum network-induced delay that the networked control system can tolerate;
- 2) one can disclose the relationship between the threshold parameter σ and the admissible minimum H_∞ performance level γ . Simulation shows that, for given $h, \bar{\tau}$, the smaller the threshold, the better the performance;
- 3) if one sets the weighting matrix Ω to be identity, for given scalars $h, \bar{\tau}$, and γ , the threshold parameter σ and the controller gain can be designed jointly based on solutions to linear matrix inequalities.

It is clear that the LKF $V_n(t)$ does not fully utilize a saw-tooth structure characteristic of an artificial delay $d(t)$, that is, the feature of $\dot{d}(t) = 1$, *a.e. (almost everywhere)* is ignored, which definitely leads to conservative results [59]. For this reason, a time-dependent LKF method is proposed to reduce the conservatism of the obtained results [60], [61]. Consequently, it is suggested in [42] to employ the following time-dependent LKF candidate as

$$V(t) = V_n(t) + V_d(t) \quad (15)$$

where $V_n(t)$ is defined in (15) and

$$V_d(t) := (\tau_M - d(t))[x(t) - x(t - d(t))]^T Z_1[x(t) - x(t - d(t))] + (\tau_M - d(t)) \int_{t-d(t)}^t \dot{x}^T(s) Z_2 \dot{x}(s) ds.$$

By the definition of $d(t)$ in (11), the aforementioned LKF $V_d(t)$ is discontinuous on $[t_0, \infty)$. By Lemma 1 in [60] or [61], $V_d(t)$ should not increase along jumping points so that the stability of the closed-loop system can be ensured. *Unfortunately, this condition is not satisfied.* To show that, we first notice that from (6) the jumping points are given by $\{s_k^i | k = 1, 2, \dots, \infty, i = 0, 1, \dots, \rho_{t_k}\}$, where $s_k^0 := t_k h + \tau_{t_k}$, $s_k^i := t_k h + h + \bar{\tau}$ ($i = 1, \dots, \rho_{t_k} - 1$), and $s_k^{\rho_{t_k}} := t_{k+1} h + \tau_{t_k}$. Then, we now consider the points s_k^i ($i = 2, \dots, \rho_{t_k} - 1$). From (11), it is clear to verify that $d(s_k^i) = \bar{\tau}$, $\lim_{t \rightarrow (s_k^i)^-} d(t) = h + \bar{\tau}$, which follows that

$$\begin{aligned} V_d(s_k^i) &= h[x(s_k^i) - x(s_k^i - \bar{\tau})]^T Z_1[x(s_k^i) - x(s_k^i - \bar{\tau})] \\ &\quad + h \int_{(s_k^i - \bar{\tau})}^{s_k^i} \dot{x}^T(s) Z_2 \dot{x}(s) ds \\ \lim_{t \rightarrow (s_k^i)^-} V_d(t) &= 0. \end{aligned}$$

That is, at the jumping point s_k^i , $i \in \{2, \dots, \rho_{t_k} - 1\}$, one has $V_d(s_k^i) \geq \lim_{t \rightarrow (s_k^i)^-} V_d(t)$, which implies that, even though the derivative of the LKF $V(t)$ in (16) along with the trajectory of the closed-loop system involved is negative almost everywhere, it is possible that the system is unstable. Therefore, one can *not* ensure the stability of the closed-loop system if the time-dependent LKF (16) is employed. How to deal with this issue remains a question.

B. Sampled-Data-Based Event-Triggered DOF Control

By the sampled-data-based ETS, DOF control is investigated in [48] and [62]. Suppose that the state of the physical plant in Fig. 1 is not available. Then, the system measurements can be used to design DOF controllers. Following an idea of [42], a time-delay system model is derived for event-triggered DOF control. Let the system measurement $y(t)$ be $Cx(t)$, where C is a known real matrix. Then, the event-triggered condition can be defined as

$$\|\Omega^{\frac{1}{2}}[y(t_k h + j h) - y(t_k h)]\|_2 \leq \varepsilon \|\Omega^{\frac{1}{2}} y(t_k h)\|_2 \quad (16)$$

where $\varepsilon > 0$ is a threshold parameter and $\Omega > 0$ is a weighting matrix. Similar to (11), we introduce two piecewise functions $d(t)$ and $e(t)$ as

$$\begin{cases} d(t) = t - t_k h - (j - 1)h \\ e(t) = y(t_k h) - y(t_k h + (j - 1)h), \quad t \in \Pi_j. \end{cases}$$

A stabilizing DOF controller can be designed of the following form:

$$\begin{cases} \dot{x}_c(t) = A_K x_c(t) + B_K x_c(t - d(t)) + C_K y(t_k h) \\ u(t) = D_K x_c(t), \quad t_k h \leq t < t_{k+1} h \end{cases} \quad (17)$$

where A_K, B_K, C_K , and D_K are constant real matrices to be determined.

It is worth pointing out that the DOF controller (18) is a *memory* controller due to the term $B_K x_c(t - d(t))$. In general, a memoryless controller can achieve better performance than the memoryless controller. Another advantage of introducing the memory controller (18) is that it is easy to be designed in terms of linear matrix inequalities.

Set $\tilde{x}(t) := \text{col}\{x(t), x_c(t)\}$. Then, the resultant closed-loop system of the plant (1) with the controller (18) is given as

$$\dot{\tilde{x}}(t) = A_0 \tilde{x}(t) + A_1 \tilde{x}(t - d(t)) + A_2 e(t), \quad t \in \Pi_j \quad (18)$$

where $A_0 := \begin{bmatrix} A & BD_K \\ 0 & A_K \end{bmatrix}$, $A_1 := \begin{bmatrix} 0 & 0 \\ C_K C & B_K \end{bmatrix}$, $A_2 := \begin{bmatrix} 0 \\ C_K \end{bmatrix}$, and $e(t)$ satisfies the following constraint for $t \in \Pi_j$

$$\begin{aligned} e^T(t) \Omega e(t) &\leq \varepsilon^2 [e(t) + CE\tilde{x}(t - \eta(t))]^T \\ &\quad \times \Omega [e(t) + CE\tilde{x}(t - \eta(t))] \end{aligned} \quad (19)$$

where $E = [I \ 0]$. The system (19) subject to (20) delivers a time-delay system model for event-triggered DOF control. By choosing a proper discontinuous LKF, event-triggered dissipative DOF control is studied in [62], which includes event-triggered passive control, H_∞ control, and mixed H_∞ and passive control as its special cases. Specifically, the aforementioned memory controller (18) can be designed as

$$\begin{cases} A_K = (W_4 - YAX - YBW_1)(I - YX)^{-1} \\ B_K = (W_3 - W_2CX)(I - YX)^{-1} \\ C_K = W_2, \quad D_K = W_1(I - YX)^{-1} \end{cases} \quad (20)$$

where $(W_1, W_2, W_3, W_4, X, Y)$ is a solution to some linear matrix inequalities. It should be mentioned that, if we set $B_K = 0$ (in this situation, the controller (18) is a memoryless controller), it is clear from (21) that $W_3 = W_2CX$, which is an equality constrain on matrix variables W_2, W_3 , and X . With such a constraint, the related matrix inequalities are no longer linear, which means that it is quite difficult to obtain a solution of matrix variables $(W_1, W_2, W_3, W_4, X, Y)$. Thus, the aforementioned memory controller is much easier to be designed than the corresponding memoryless controller.

The DOF control under a memoryless DOF controller is studied in [53] using event-triggered quantization schemes. Whether or not the sampled signals should be quantized is determined by an ETS or a sampled-data-based ETS. A hybrid system approach and a time-delay system approach are developed to deal with stability of the closed-loop system. Moreover, under either the ETS or the sampled-data-based ETS, suitable DOF controllers can be designed in terms of solutions to linear matrix inequalities if the system output matrix is of full-row rank.

IV. NETWORK-BASED H_∞ FILTERING USING THE SAMPLED-DATA-BASED ETS

The filtering (or called state estimation) problem is one of fundamental problems in the field of control. For a dynamic system in the presence of process noise, the objective of the filtering problem is to estimate system states (or a linear

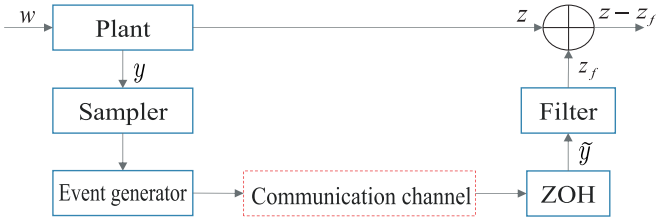


Fig. 2. Simple diagram for network-based H_∞ filtering.

combination of them) using noisy output measurements. The celebrated Kalman filtering approach provides a recursive algorithm to minimize the variance of the state estimation error when the noise is random with known statistical properties or has a known power spectral density. Kalman filtering approaches have found a wide range of applications in industrial control [63], such as power station control systems, dynamic positioning systems, high-performance drive systems, etc.; see, e.g., special issues in [64] and [65].

However, Kalman filtering approaches are sensitive to uncertainties in the exogenous noise signals. If statistical properties of the noise are unknown, Kalman filtering approaches inapplicable. In this case, H_∞ filtering can be applied. The H_∞ filtering approach was first considered in [66] (in the scalar case) and in [67] (in the multivariable case) using polynomial techniques. The H_∞ filtering approach aims to minimize the energy of the estimation error for the worst possible bounded energy noise. In comparison with Kalman filtering, a significant characteristic of H_∞ filtering is that only upper bounds on the spectral density of the input and measurement noises need to be known. As a result, H_∞ filtering has attracted considerable attention since the late of 1980s [68]–[73].

A. Description of Event-Based H_∞ Filtering

Fig. 2 shows a simple diagram for event-based H_∞ filtering of a networked system, where the physical plant is the linear system described by

$$\begin{cases} \dot{x}(t) = Ax(t) + Bw(t), & x(0) = x_0 \\ z(t) = Lx(t) \\ y(t) = Cx(t) + Dw(t) \end{cases} \quad (21)$$

where $x(t)$ is the system state vector, $w(t)$ the noise vector belonging to $\mathcal{L}_2[0, \infty)$, $y(t)$ the measurement vector, and $z(t)$ the controlled output vector to be estimated. The filter to be designed is of the following form:

$$\begin{cases} \dot{x}_f(t) = A_f x_f(t) + B_f \tilde{y}(t), & x_f(0) = 0 \\ z_f(t) = C_f x_f(t) + D_f \tilde{y}(t) \end{cases} \quad (22)$$

where $x_f(t)$ is the filter state and $z_f(t)$ is the estimation of $z(t)$. If $D_f = 0$, the filter is called a strictly proper filter. The event-based H_∞ filtering problem can be described as the following event-based γ -suboptimal H_∞ filtering problem.

Event-based γ -suboptimal H_∞ filtering problem: Under an ETC, determine the filter gain matrices A_f , B_f , C_f , and D_f such that for $w \in \mathcal{L}_2/\{0\}$ the L_2 -induced gain from the noise

signal w to the estimation error $z - z_f$ is less than a prescribed H_∞ level $\gamma > 0$, i.e.

$$\frac{\|z - z_f\|_2^2}{\|w\|_2^2} \leq \gamma^2. \quad (23)$$

It is worth pointing out that the aforementioned description is based on an assumption that the system initial value x_0 is known (if x_0 is known, we can set $x_0 = 0$ without loss of generality because the physical plant (22) is linear). If x_0 is not known exactly, but in the neighborhood of $x_0 = 0$ with 0 as the best *a priori* estimation of x_0 , the denominator $\|w\|_2^2$ in (24) should be replaced with $\|w\|_2^2 + x_0^T R x_0$, where $R = R^T > 0$ is a weighting matrix, indicating a measure of relative importance of the uncertainty. The analysis approach to the H_∞ filtering under zero initial conditions is similar to that under unknown initial conditions. The only difference is that the latter needs to choose the weighting matrix R such that the confidence in the *a priori* knowledge of the initial state can be reflected. Therefore, most of results have been derived under zero initial conditions for simplicity of presentation. In addition, if setting the time $t \in [0, T]$ in the physical plant (22), where $T < \infty$, the corresponding H_∞ filtering is called finite-horizon H_∞ filtering.

B. Time-Delay System Approach to Event-Based H_∞ Filtering

Recently, based on an idea of the sampled-data-based ETS, event-based H_∞ filtering has gained increasing attention and a number of results have been reported in the literature. To mention a few, event-based H_∞ filtering is studied for networked systems with communication delays [74], for sampled-data systems [20], for complex networks with mixed time delays [75], and for discrete-time linear systems [76]; event-triggered generalized dissipativity filtering for neural networks is investigated in [77]. A common feature of the aforementioned results is that a time-delay system approach is proposed to design suitable event-triggered H_∞ filters.

Following the line in [48], it is not difficult to establish a time-delay model for the filtering error system. Since the system measurement $y(t)$ is available, the event-triggered condition can be defined in (17), based on which, a time-delay model for the resultant filtering error system can be given by

$$\begin{cases} \dot{\tilde{x}}(t) = \tilde{A}_0 \tilde{x}(t) + \tilde{A}_1 \tilde{x}(t - d(t)) + \tilde{B}_1 \tilde{w}(t) + \tilde{B}_2 e(t) \\ \mathcal{E}(t) = \tilde{L}_0 \tilde{x}(t) + \tilde{L}_1 \tilde{x}(t - d(t)) + \tilde{D}_1 \tilde{w}(t) + \tilde{D}_2 e(t) \\ \tilde{x}(\theta) = \phi(\theta) = \text{col}\{x_0, 0\}, & t \in \Pi_j \end{cases} \quad (24)$$

where $\tilde{x} := \text{col}\{x(t), x_f(t)\}$, $\mathcal{E}(t) := z(t) - z_f(t)$, $\tilde{A}_0 := \text{diag}\{A, A_f\}$, $\tilde{A}_1 := \text{col}\{0, B_f C\}$, $\tilde{B}_1 := \text{diag}\{B, B_f D\}$, $\tilde{B}_2 := \text{col}\{0, B_f\}$, $\tilde{L}_0 := [L - C_f]$, $\tilde{L}_1 := -D_f C$, $\tilde{D}_1 := [0 - D_D]$, and $\tilde{D}_2 := -D_f$.

With the aforementioned time-delay model (25), the LKF method can be employed to solve the problem of event-based H_∞ filtering. In [74], by choosing a continuous LKF candidate

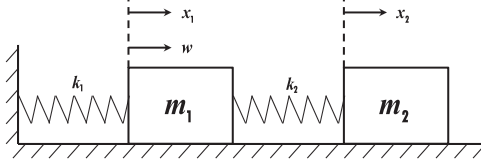


Fig. 3. Mass-spring system.

as

$$\begin{aligned} \tilde{V}_n(t) = & \tilde{x}^T P \tilde{x}(t) + \int_{t-h-\bar{\tau}}^t x^T(s) Q x(s) ds \\ & + \int_{-h-\bar{\tau}}^0 \int_{t+\theta}^t \dot{x}^T(s) R \dot{x}(s) ds d\theta \end{aligned} \quad (25)$$

some sufficient conditions are derived on the existence of event-triggered H_∞ filters. However, these conditions are of somewhat conservatism since the feature of $\dot{d}(t) = 1, a.e.$ is not considered. Suppose that the network-induced delay τ_{t_k} is a constant τ , i.e., $\tau_{t_k} \equiv \tau > 0$. Then, some less conservative sufficient conditions are obtained in [20] by choosing a discontinuous LKF as $\tilde{V}(t) = \tilde{V}_n(t) + \tilde{V}_d(t)$, where

$$\begin{aligned} \tilde{V}_d(t) := & h^2 \int_{(t_k+j-1)h}^t \dot{x}^T(s) R \dot{x}(s) ds \\ & - \frac{\pi^2}{4} \int_{(t_k+j-1)h}^{t-\tau} \varpi^T(s) R \varpi(s) ds, \quad t \in \Pi_j \end{aligned}$$

with $\varpi(s) := x(s) - x((t_k + j - 1)h)$. By using Wirtinger inequality, some bounded real lemma is derived for the event-based filtering error system (25), based on which, an algorithm is presented to design both the event threshold parameter ε and the H_∞ filter parameters A_f, B_f, C_f , and D_f , simultaneously, in terms of solutions to a set of linear matrix inequalities.

C. Application to a Mechanical System With Two Masses and Two Springs

In this subsection, we take an example to show the effectiveness of the event-based H_∞ filtering approach. A mechanical system with two masses and two springs is shown in Fig. 3, where x_1 and x_2 are the positions of the masses m_1 and m_2 , respectively; and k_1 and k_2 are spring constants, see [78] for more details. The motion equations of x_1 and x_2 can be given by

$$\begin{cases} \ddot{x}_1 = -\frac{k_1 + k_2}{m_1} x_1 + \frac{k_2}{m_1} x_2 - \frac{c}{m_1} \dot{x}_1 + \frac{1}{m_1} w \\ \ddot{x}_2 = \frac{k_2}{m_2} x_1 - \frac{k_2}{m_2} x_2 - \frac{c}{m_2} \dot{x}_2 \end{cases}$$

where c denotes the viscous friction coefficient between the masses and the horizontal surface. Now we assume that the measurements of x_1 and x_2 are disturbed by w , which is supposed to be

$$w(t) = \begin{cases} 1, & 1s \leq t \leq 3s \\ -1, & 5s \leq t \leq 8s \\ 0, & \text{otherwise.} \end{cases}$$

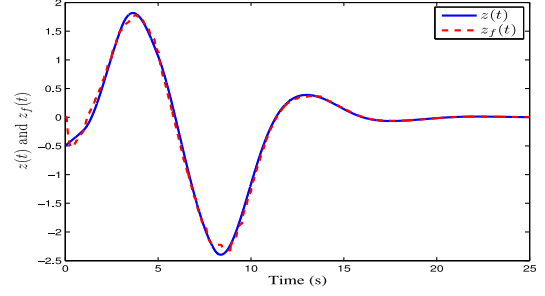


Fig. 4. Controlled output $z(t)$ and its estimate $z_f(t)$ under the event-triggering scheme.

Our objective is to design an event-based H_∞ filter to estimate $x_1 + x_2$. Let $x := \text{col}\{x_1, x_2, \dot{x}_1, \dot{x}_2\}$ and $m_1 = 1, m_2 = 0.5, k_1 = k_2 = 1$, and $c = 0.5$. Then, a state-space realization of this plant can be described by (22) with

$$\begin{aligned} A &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 1 & -0.5 & 0 \\ 2 & -2 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \\ C &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}, \quad L = [1 \ 1 \ 0 \ 0]. \end{aligned}$$

Suppose the admissible minimum H_∞ performance $\gamma = 0.8$, the sampling period $h = 0.08$ s and the transmission delay $\tau = 0.1$ s. Then, employing Algorithm 1 [20], the event parameters (ε, Ω) and a strictly proper filter (A_f, B_f, C_f) can be codesigned, which are given by $\varepsilon = 0.1313$ and

$$\begin{aligned} \Omega &= \begin{bmatrix} 5.7007 & -0.1704 \\ -0.1704 & 2.5549 \end{bmatrix} \\ A_f &= \begin{bmatrix} -7.8770 & -2.0059 & 0.9983 & 0.0037 \\ -4.7156 & -4.7380 & 0.0015 & 1.0004 \\ -8.4649 & 6.4694 & 0.9874 & 0.3521 \\ 1.1548 & 0.5963 & -0.0014 & -1.0002 \end{bmatrix} \\ B_f &= \begin{bmatrix} -7.6013 & -1.9347 \\ -4.5395 & -4.5834 \\ -5.5992 & 4.3988 \\ -0.8270 & 2.5168 \end{bmatrix}, \quad C_f = [-1 \ -1 \ 0 \ 0]. \end{aligned}$$

Connecting the obtained filter with the plant, and taking the event-triggered condition (17) into account, the curves of the signal $z(t)$ and the estimated signal $z_f(t)$ are depicted in Fig. 4, where the initial condition of the plant is taken as $x_0 = [-0.5 \ 0 \ 0.2 \ 0.1]^T$. Moreover, we calculate the communication resource utilization during the time period $[0, 25]$ s. During this period, the number of sampled data packets is 312 while only 147 packets (47.12% of 312) are transmitted to the filter due to the introduction of the ETS, leading to a 52.88% saving of communication resources. The release instants and the time distance between any two adjoint release instants are illustrated in Fig. 5. On the other hand, we reconsider the aforementioned filter in the purely time-triggered scenario, which means that all the sampled data packets should be transmitted to the filter. In this situation, the controlled output $z(t)$

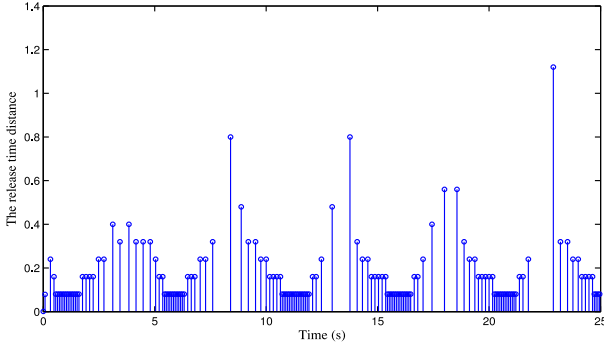


Fig. 5. Release instants and time intervals.

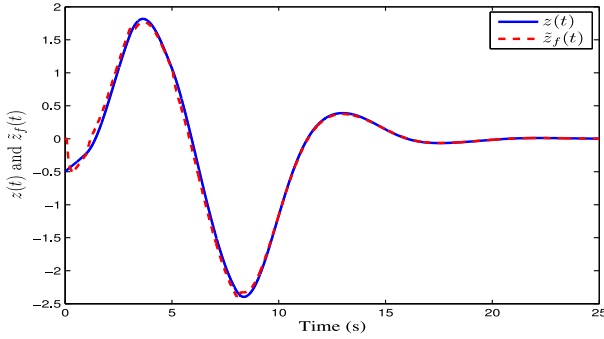


Fig. 6. Controlled output $z(t)$ and its estimate $\tilde{z}_f(t)$ under the time-triggered scheme.

and its estimation denoted by $\tilde{z}_f(t)$ are shown in Fig. 6. From Figs. 4 and 6, one can see clearly that both $z_f(t)$ and $\tilde{z}_f(t)$ are good estimations on the signal $z(t)$. However, under the ETS, the communication resources can be saved by 52.88% of those under the time-triggered scheme.

It is worth pointing out that simulation results in [20] also gain an insight into the relationship between time-based and event-based H_∞ filtering. It is found that one cannot claim that “by choosing a larger sampling period for time-based H_∞ filtering one would always get better results than those using the sampled-data-based ETS.”

V. SAMPLED-DATA-BASED EVENT-TRIGGERED CONSENSUS FOR DISTRIBUTED MASS

Since the pioneer work of Olfati-Saber and Murray (2004) [79], a consensus problem of distributed MASs has become a very attractive research topic due to its wide applications in cooperative control of mobile robots, formation control of unmanned vehicles, and so on [80], [81].

A. Distributed Event-Triggered Transmission Strategy

Consider a distributed MAS with N agents whose dynamics are described as

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \quad x_i(0) = x_i^0, \quad (i = 1, \dots, N). \quad (26)$$

The communication topology is dominated by a directed weighted graph $\mathcal{G} = \{\Delta, \mathcal{E}, \mathcal{W}\}$, where Δ and $\mathcal{E}$ are the sets of agents and edges, respectively, and $\mathcal{W} = (w_{ij})_{N \times N}$ a weighted

adjacent matrix. The consensus problem of system (27) is to design a suitable consensus protocol such that a consensus can be asymptotically reached, i.e., $\lim_{t \rightarrow \infty} \|x_i(t) - x_j(t)\| = 0$ for any initial condition x_i^0 and $i, j = 1, \dots, N$. In [79], two types of consensus protocols are proposed for two different cases: 1) for fixed or switching topology and zero communication time-delay, a consensus protocol can be given as

$$u_i(t) = K \sum_{v_i \in N_i} w_{ij}(x_j(t) - x_i(t)) \quad (27)$$

where K is the controller gain to be designed and N_i denotes the set of neighbors of agent v_i ; and 2) for fixed topology and communication time delay τ_{ij} corresponding to the edge $\varepsilon_{ij} \in \mathcal{E}$, a consensus protocol can be of the following form:

$$u_i(t) = K \sum_{v_i \in N_i} w_{ij}(x_j(t - \tau_{ij}) - x_i(t - \tau_{ij})). \quad (28)$$

Clearly, there is an ideal assumption on the aforementioned protocols that *continuous-time signals* should be transmitted from neighboring agents v_j to v_i , which is unpractical if digital networks are used.

When data transmission between agents is in the form of digital signals, following an idea of the sampled-data-based ETS, a distributed event-triggered transmission strategy is proposed in [47] to save limited communication resources while a consensus is asymptotically achieved. It is assumed that the states of all agents are sampled with a period $h > 0$. Whether or not the sampled signal of agent i is broadcast depends on whenever its predefined event-triggered condition is violated at the sampling instant kh ($k \in \mathbb{Z}^+$). Denote by $t_{m_i}^i h$, the last event time instant indicating that the sampled signal of agent i is broadcast. Then, the next event time instant is determined by the first time instant when the following event-triggered condition is violated:

$$\chi_i^T(t_{m_i}^i h + l_i h) \Phi \chi_i(t_{m_i}^i h + l_i h) \leq \sigma_i \psi_i^T(t_{m_i}^i h) \Phi \psi_i(t_{m_i}^i h) \quad (29)$$

where σ_i is the threshold parameter and $\Phi > 0$ a weighting matrix, and

$$\chi_i(t_{m_i}^i h + l_i h) = x_i(t_{m_i}^i h + l_i h) - x_i(t_{m_i}^i h)$$

$$\psi_i(t_{m_i}^i h) = \sum_{j=1}^N w_{ij}[x_i(t_{m_i}^i h) - x_j(t_{m_j}^j h)].$$

B. Time-Delay System Model for Consensus of MASs

Suppose that the effects of network-induced delays and packets dropouts can be neglectable. Then, one can seek the following consensus protocol based on (28)

$$u_i(t) = -K \sum_{v_i \in N_i} w_{ij}[x_i(t_{m_i}^i h) - x_j(t_{m_j}^j h)] \quad (30)$$

where $t \in [t_{m_i}^i h, t_{m_i+1}^i h) = \cup_{k=t_{m_i}^i}^{t_{m_i+1}^i-1} [kh, (k+1)h)$. In order to deal with the consensus problem, a time-delay system method is proposed in [47].

First, introduce an error vector as $e_i(kh) = x_i(kh) - x_i(t_{m_i}^i h)$. Then, the event-triggered condition (30) can be

rewritten as $e_i^T(kh)\Phi e_i(kh) \leq \sigma_i \mu_i(kh)^T \Phi \mu_i(kh)$, where $\mu_i(kh) = \sum_{j=1}^N [e_j(kh) - e_i(kh) + x_i(kh) - x_j(kh)]$.

Second, following a line in [82], the consensus problem can be converted into the asymptotic stability problem. Introduce a transformation $z_i(t) = x_1(t) - x_i(t)$ and denote $z(t) := \text{col}\{z_1(t), z_2(t), \dots, z_N(t)\}$. Then, the closed-loop system associated with (27) and (31) can be given by

$$\dot{z}(t) = (I_{N-1} \otimes A)z(t) - (\mathcal{L} \otimes BK)z(kh) + (\mathcal{L} \otimes BK)\mathcal{E}(kh) \\ t \in [kh, (k+1)h) \quad (31)$$

where $\mathcal{E}(kh) := \text{col}\{e_1(kh) - e_2(kh), e_1(kh) - e_3(kh), \dots, e_1(kh) - e_N(kh)\}$ and $\mathcal{L} := [\mathbf{1} \quad -I_{N-1}]L[\mathbf{0} \quad -I_{N-1}]^T$ with L being the Laplacian matrix of the directed graph $\mathcal{G}$. Thus, the consensus of the MAS (27) is converted into asymptotic stability of the system (32).

Third, set $\tau(t) = t - kh$, $kh \leq t < (k+1)h$. Then, a time-delay model is obtained as

$$\dot{z}(t) = (I_{N-1} \otimes A)z(t) - (\mathcal{L} \otimes BK)z(t - \tau(t)) \\ + (\mathcal{L} \otimes BK)\mathcal{E}(t - \tau(t)), \quad t \in [kh, (k+1)h). \quad (32)$$

By employing the Lyapunov stability theory, the consensus problem can be solved in terms of solutions to a set of linear matrix inequalities (see [47] in detail).

The aforementioned time-delay system approach is applied to solve the problem of finite-time distributed event-triggered consensus control for MASs in [83], and is also extended to discrete-time MASs with switching topology in [84].

C. Some Remarks

From the aforementioned analysis, one can see clearly that an idea of the sampled-data-based ETS is successfully applied to event-triggered consensus of MASs [47], and a time-delay system approach is proposed to design suitable event-triggered consensus protocols such that communication resources can be significantly saved while a consensus of the closed-loop system can be ensured. However, there is much room for improvement since the obtained results are based on a couple of ideal assumptions and some issues need to be further discussed.

- 1) The first ideal assumption is that there are no transmission delays when signals are transmitted between agents through communication networks, under which, consequently, the consensus protocol of form (28) is employed. In practice, transmission delays are unavoidable due to the limited bandwidth of communication networks. Usually, transmission delays are regarded as a major factor of degrading the performance of an MAS. Thus, transmission delays, together with the consensus protocol of form (29), should be taken into account.
- 2) The second ideal assumption is that signals from different agents are sampled synchronically with the same period h . This assumption requires that all the agents check their respective event-triggered condition always at the same time instants event though they are located in different places. As a result, such a phenomenon occurs: before the sampled data packet from agent j is triggered, it is

used for the neighboring agent i ($i \neq j$) to determine whether or not to trigger the current sampled data packet from agent i , which is impossible in practice.

- 3) How to define a proper event-triggered condition for a distributed MAS needs to be further studied. The event-triggered condition defined in (30) is convenient to use in the consensus analysis, but its physical meanings are not clear. In fact, one cannot explain why the event-triggered condition should be defined as that in (30) from the physical point of view. Moreover, the weighting matrix Φ is not necessarily the same for all the agents, which usually leads to conservative results.
- 4) In a directed graph, event-triggered conditions should take the weighting adjacency matrix into consideration. The weighting adjacency matrix indicates what edges are important and what edges are not so important. Thus, for the important edges, corresponding sampled-data packets should be transmitted more than those for unimportant edges so that a consensus can be reached as soon as possible.

VI. APPLICATIONS OF SAMPLED-DATA-BASED ETC AND FILTERING

Due to the advantages that the sampled-data-based ETS offers, the sampled-data-based ETS has been applied to systems, such as offshore structures, power systems, distributed filtering, and complex networks, and one can refer to [75] and [85]–[90].

- 1) In [85], H_∞ reliable control for offshore structures is investigated using the sampled-data-based ETS, under which both control cost and communication resources can be saved significantly while certain control performance of the offshore structures can be maintained.
- 2) In [86], an event-triggered transmission optimal scheme is employed for economic dispatch to reduce information exchange requirements in smart grids, which provides a safety technique that all the generators are running below their capacity with lower communication requirements.
- 3) In [87], load frequency control is concerned with for power systems with communication delays via a sampled-data-based ETS. Simulation results show that this scheme can reduce the amount of communications required for a prescribed performance level.
- 4) In [88], distributed H_∞ filtering over sensor networks is studied using the sampled-date-based ETS. The measurements of the physical plant are monitored by a sensor network with a certain topology. An event-triggered distributed filtering approach is proposed to design suitable distributed H_∞ filters such that a consensus state estimation of the physical plant can be achieved with a significant reduction of limited communication resources, which is verified through a quarter-car suspension model.

VII. CONCLUSION AND FUTURE RESEARCH

An overview on sampled-data-based ETC and filtering for networked systems has been provided. Under the sampled-data-based ETS, a positive minimum inter-event time is

ensured, and feedback controllers and event-triggered parameters can be jointly designed. A deep investigation of the sampled-data-based ETS has been made. Second, recent results on sampled-data-based ETC, sampled-data-based event-based H_∞ filtering, and sampled-data-based event-triggered consensus for distributed MASs have been reviewed. These results have been deeply analyzed and some remarks/comments have been given. Some applications of sampled-data-based ETC/filtering have been briefly reviewed.

Although a number of notable results on sampled-data-based ETC and filtering have been reported in the literature, the following problems are still challenging, which need to be considered in the near future research.

- 1) For a given threshold parameter in (2), it is possible that the sampled-data-based event-triggered condition will *not* be always violated after some time instant say t_f , especially when the system approaches its steady state. An efficient method that overcomes this limitation is to define the following event-triggered condition as

$$\sum_j \|y(t_k h + j h) - y(t_k h)\|_2 \leq \varepsilon \|y(t_k h)\|_2. \quad (33)$$

When the error $\|y(t_k h + j h) - y(t_k h)\|_2$ is accumulated to a certain extent, the condition (34) will be violated, which means that a new event will be triggered. However, no results have been reported yet on how the condition (34) works for ETC and filtering.

- 2) Under the sampled-data-based ETS, how to deal with packet dropouts is an important issue due to that packet dropout phenomena are common when sampled-data packets are transmitted through a communication network. In [91], one way to handle packet dropouts is suggested. If a sampled-data packet is lost, one can increase the threshold σ in the event-triggered condition (2) to some extent such that the lost packet will not be released. However, to what extent the threshold parameter σ should be increased such that the lost packet exactly satisfies the modified event-triggered condition, it is difficult to be determined in a practical networked control system. Up to date, no efficient results have been obtained to deal with packet dropouts in ETC.
- 3) The sampled-data-based ETS requires that system measurements should be sampled periodically. If the sampling is stochastic or time-varying, the time-delay system model (14) is no longer available. How to establish an effective mathematical model to suit for the stochastic or time-varying sampling such that both controllers and event-triggered threshold parameters can be jointly designed is also a challenging problem.
- 4) In distributed MASs, a sampled-data-based ETS can be employed to deal with several issues, such as distributed consensus, distributed estimation, cooperative control, and leaderless and leader-following coordination [92]–[94]. However, up to date, few results have been reported in the literature. Moreover, when the sampled-data-based ETS is applied, it is significant in the sense of saving limited communication resources, but it is

challenging if the following situations are taken into account (not necessarily simultaneously).

- a) Each agent samples its signals in a different period. In this case, every agent will check its event-triggered condition at different time instants. As a result, it is not an easy task to establish an effective mathematical model for the closed-loop system.
 - b) The communication topology is time varying, which often occurs if agents are unmanned air vehicles. In this situation, how to define effective event-triggered conditions needs to be deeply investigated.
 - c) Signals are transmitted subject to network-induced delays and packet dropouts. Network-induced delays and packet dropouts may change a communication topology of a distributed system, which makes it quite difficult to design suitable consensus and coordination algorithms even if the sampled-data-based ETS is not applied.
- 5) Dynamic event-triggered mechanisms are proposed recently in [25] and [29]. Compared with static event-triggered mechanisms, dynamic event-triggered mechanisms can enlarge the minimum inter-event time even close to the maximum allowable transmission interval [29]. Dynamic event-triggered mechanisms are expected to become a hot research topic in the coming years. How to apply an idea of dynamic event-triggered mechanisms to the sampled-data-based ETS is significant in theory and in practice.
 - 6) A quasi-synchronization problem in leader-follower heterogeneous dynamic networks is investigated using distributed impulsive control method in [95], where a distributed impulsive protocol based on pinning controller is designed. There is an implicit assumption that all the nodes' impulses should be synchronized in time, which is quite difficult in practice. How to apply an idea of the sampled-data-based ETS to the proposed control scheme such that this assumption can be removed is still challenging.

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